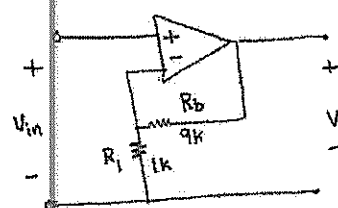
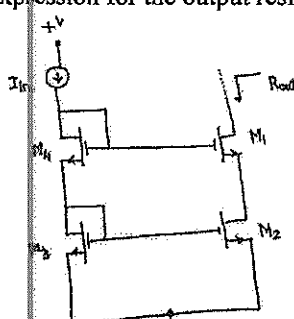
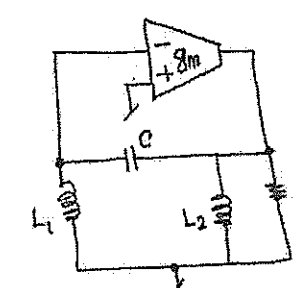
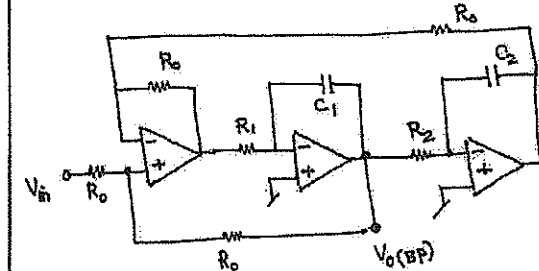


(c)	For the non-inverting amplifier shown in Fig.3, the open loop gain 'A' of the op-amp is $A \cong \frac{\omega_t}{s}$ where ω_t is the gain bandwidth product equal to $2\pi \cdot 10^6$. Determine the bandwidth of the close-loop amplifier by deriving the formula used.	5	CO3
 Fig.3			
4.	Attempt any two parts		
(a)	Compare the simple, Widler and Wilson current mirrors in terms of the error in the current gain and output resistances.	5	CO4
(b)	For the MOS cascode current mirror shown in Fig.4, draw the small signal model and hence, determine the approximate expression for the output resistance.	5	CO4
 Fig.4			

(c)	Explain the operation and transfer characteristics of push-pull amplifier.	5	CO4
5.	Attempt any two parts		
(a)	For the Hartley oscillator shown in Fig.5, derive the expressions for the condition of oscillation and the frequency of oscillation.	5	CO5
 Fig.5			
(b)	Draw the circuit diagram of astable multivibrator using either op-amp or 555-IC timer and explain its operation. Also, determine the expression for the frequency of the generated square wave.	5	CO5
(c)	For the KHN filter shown in Fig.6 determine the centre frequency of Band Pass Filter (BPF), given that $R_o = 10k\Omega$, $R_1 = R_2 = 100k\Omega$ and $C_1 = C_2 = 1nF$.	5	CO5
 Fig.6			